

SECTION A — CONCEPT APPLICATION

A1. Compatible Hardware Selection

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

Answer: First, identify the workstation requirements, including the type of office applications, required RAM, storage capacity, performance needs, and budget. Select a motherboard and CPU that use a compatible CPU socket and are supported by the motherboard chipset and BIOS/UEFI. Select the correct DDR RAM generation, capacity, speed, and configuration supported by the motherboard. For storage, confirm that the motherboard supports the required SATA ports or M.2/NVMe interface. Finally, check the motherboard manual, CPU support list, RAM compatibility information, physical dimensions, and PSU requirements to confirm that all components work together.

A2. HDD vs SSD vs NVMe

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Answer: An HDD stores data on spinning magnetic disks and normally offers high capacity at lower cost, but it has moving parts and slower access. A SATA SSD uses flash memory, has no moving parts, and provides much faster access, quieter operation, and better resistance to physical shock. An NVMe SSD is also flash-based but normally communicates over PCIe using the NVMe protocol, giving higher throughput and lower latency than SATA storage. For office workstations, an SSD is well suited because it provides fast booting, quick application loading, and reliable everyday performance. NVMe is an option when the motherboard supports it and higher storage performance is required.

A3. GPU and Cooling

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under several hours of mildly graphics-intensive use.

Answer: The GPU processes graphics and can accelerate graphics-related tasks in design applications. During sustained workloads, the CPU and GPU produce heat. The cooling system, including heatsinks, fans, case airflow and, where applicable, liquid cooling, removes this heat and helps maintain safe operating temperatures. Effective cooling reduces thermal throttling and helps prevent instability, unexpected shutdowns, and excessive component temperatures during long work sessions.

A4. Hardware Compatibility

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Answer: Hardware compatibility is important because components must communicate correctly, be physically installable, and receive the correct power and interfaces. Examples include CPU socket compatibility, supported RAM generation, supported storage interface, GPU expansion-slot and clearance requirements, and PSU power/connectors. Incompatible hardware may not start, may not be detected, may operate incorrectly, or may cause instability. Physical incompatibility can also prevent a component from being installed safely.

A5. PSU Selection and Safety

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

Answer: Estimate the power requirements of the CPU, GPU, storage, fans, and other components, then select a quality PSU with sufficient continuous wattage and reasonable headroom. Confirm that it has the required motherboard, CPU/EPS, GPU PCIe, and storage connectors. Before powering on, disconnect AC power while working inside the case, use ESD precautions, inspect cables and connectors, and make sure the motherboard 24-pin and CPU power connectors are fully seated. Confirm GPU and storage connections and check for loose screws, incorrect connections, or anything that could cause a short circuit.

A6. POST Verification

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

Answer: After powering on, check that the system starts normally without repeated restart cycles or diagnostic errors. Confirm that the display appears and that BIOS/UEFI detects the CPU, installed RAM, and storage device. Check diagnostic LEDs or beep codes if present. If POST fails, switch off power and inspect PSU connections, CPU power, RAM seating, GPU seating, display connections, and CPU/motherboard compatibility. Reseat RAM or test one module at a time if necessary, and clear CMOS according to the motherboard instructions before retesting.

SECTION B — PRACTICAL CODING TASKS

B1. Task 1 — Component Identification & Functions

Task requirement: Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system. Note the make/model or key specification of each component where visible. Present the findings as a labeled component list or table.

Answer / procedure: Use the following table while inspecting the actual desktop. Replace the bracketed fields with the visible make/model/specification.

Component	Function	Actual Make/Model / Specification
Motherboard	Connects and allows communication between CPU, RAM, storage, GPU, PSU and peripherals.	[Fill from hardware]
CPU	Executes instructions and performs general processing.	[Fill from hardware]
RAM	Temporary working memory for running programs and the operating system.	[Fill from hardware]
GPU	Processes graphics and provides display/graphics acceleration.	[Fill from hardware]
Storage	Stores the operating system, applications and user files permanently.	[Fill from hardware]
Cooling	Removes heat and helps maintain safe operating temperatures.	[Fill from hardware]

B2. Task 2 — Motherboard Ports & Socket Identification

Task requirement: Identify and label all major motherboard ports, expansion slots, and the CPU socket. Note which CPU families are compatible with the socket and provide a photograph or diagram with labels.

Answer / procedure: Major labels to look for include CPU socket, DIMM/RAM slots, 24-pin ATX power, CPU/EPS power, SATA ports, M.2 slots, PCIe x16 slot, other PCIe slots, USB headers, front-panel headers, front-audio header, fan headers, CMOS battery, rear USB ports, LAN port, audio ports, and video outputs where present. Record the exact CPU socket and use the motherboard manufacturer's CPU support list to identify compatible CPU families.

B3. Task 3 — RAM & Storage Installation

Task requirement: Remove and reinstall a RAM module and a storage device, observing ESD precautions, then confirm both are detected correctly.

Answer / procedure: 1. Shut down and disconnect AC power. 2. Use ESD precautions. 3. Release the RAM latches and remove the module by its edges. 4. Align the notch and reinstall it until the clips lock. 5. For SATA storage, disconnect/reconnect the data and power cables; for NVMe, remove/reinstall the retaining screw and drive. 6. Boot into BIOS/UEFI and confirm RAM capacity and storage detection. 7. Boot the operating system and verify the storage is accessible.

B4. Task 4 — Full Desktop PC Assembly

Task requirement: Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU; follow ESD/safety precautions; document each step; and confirm successful POST.

Answer / procedure: Recommended sequence: prepare the workspace and ESD protection; install CPU; install CPU cooler; install RAM; install M.2 storage if used; fit motherboard into case; install PSU; connect 24-pin and CPU power; install GPU and its required power; connect SATA storage if present; connect front-panel, USB, audio and fan headers; inspect the completed build; power on; enter BIOS/UEFI; and verify CPU, RAM and storage detection.

Evidence to attach: labeled photos/diagrams of CPU and RAM installation, storage, motherboard/case, GPU/PSU connections, completed assembly, and successful POST/BIOS screen.

SECTION C — MINI CAPSTONE PROJECT

C1. First Three Hardware Checks

Question: Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.

Answer: Check the PSU/power connections, RAM, and motherboard/CPU power and compatibility first. The PSU and its connections provide the required power; incorrectly seated RAM can prevent POST; and an incompatible CPU/motherboard combination or missing CPU power connection can prevent the system from initializing.

C2. Motherboard–CPU Compatibility

Question: Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system assembled or inspected.

Answer: The CPU must be supported by the motherboard's socket, chipset, and BIOS/UEFI version. A processor fitting physically into a socket does not by itself guarantee compatibility. For the inspected system, record the exact motherboard and CPU models and verify the processor against the motherboard manufacturer's supported-CPU list.

C3. RAM and Storage Verification

Question: Explain how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.

Answer: RAM should be correctly aligned, fully seated, and installed in the recommended slot(s). BIOS/UEFI should report the expected memory capacity. SATA storage needs correct data and power connections, while an NVMe drive must be correctly installed and secured in the M.2 slot. BIOS/UEFI should detect the storage device, and the operating system should show it correctly after boot.

C4. Hardware Installation Checklist

Question: Produce a hardware-installation checklist of at least three steps to prevent similar failures.

Answer:

- Confirm CPU/motherboard compatibility before assembly.
- Use ESD precautions and disconnect power during internal work.
- Confirm RAM is correctly oriented and fully seated.
- Confirm SATA/M.2 storage is correctly installed and connected.
- Securely connect the motherboard 24-pin and CPU power connectors.
- Check GPU seating and required power connections.
- Inspect for loose screws, incorrect standoffs and cable problems.
- Verify CPU, RAM and storage detection in BIOS/UEFI after first boot.

C5. SSD Instead of HDD

Question: Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.

Answer: SSDs provide much faster access to files and applications than traditional HDDs, which improves boot and application-loading times. They have no spinning mechanical parts, so they are quieter and generally more resistant to physical shock. For an office computer lab, SSDs provide responsive everyday performance with low mechanical maintenance. NVMe SSDs can provide additional performance when supported by the workstation hardware.

SECTION D — AI-AUGMENTED LEARNING

D1. STEP 1 — BUILD WITH AI

Question / task requirement: Use an AI tool to produce a step-by-step troubleshooting checklist for a desktop PC that fails to boot immediately after assembly. It must cover the motherboard, RAM, storage and PSU, be in priority order, and be applied to the ABC Solutions scenario.

Exact AI prompt: “Create a clear step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Put the checks in the correct priority order. Cover motherboard, RAM, storage and PSU as possible failure points. Present it as a simple on-site checklist and apply it to the ABC Solutions scenario where some newly assembled computers fail to start or show hardware detection errors.”

AI Original Checklist

- 1. Confirm AC power and PSU switch.
- 2. Check PSU cable and motherboard 24-pin connection.
- 3. Check CPU/EPS power and diagnostic LEDs/beeps.
- 4. Power off and use ESD precautions before opening the case.
- 5. Reseat RAM and test one module at a time.
- 6. Check CPU installation and cooler connection.
- 7. Check GPU seating and display cable.
- 8. Check SATA/NVMe storage and BIOS detection.
- 9. Verify CPU/motherboard compatibility and BIOS support.
- 10. Clear CMOS if required and retest POST.
- 11. After POST, check operating system and drivers.

D2. STEP 2 — TEST & DEBUG WITHOUT AI

Question / task requirement: Review the checklist without AI, identify at least one missing, incorrectly ordered, or unsafe step, and correct it.

Corrected version:

- 1. Shut down, switch off PSU, disconnect AC power and apply ESD precautions before opening the case.
- 2. Visually inspect for loose screws, incorrect standoffs, damaged parts or disconnected cables.
- 3. Reconnect power and verify the PSU switch, outlet and external cable.
- 4. Check motherboard 24-pin and CPU/EPS power connections.
- 5. Power on and observe fans, diagnostic LEDs/beeps and display output.
- 6. If POST fails, power off and reseat RAM; test one known-good module in the recommended slot.
- 7. Check CPU installation, cooler connection and motherboard/CPU compatibility.
- 8. Check GPU seating and power if a discrete GPU is installed.
- 9. Enter BIOS/UEFI and verify CPU, RAM and storage detection.
- 10. Check SATA/NVMe installation if storage is missing.
- 11. Clear CMOS according to the motherboard manual if required, then retest.
- 12. Only after hardware POST succeeds, continue to operating-system and driver checks.

D3. 3–4 Line Note

I moved the ESD and safe-power step to the beginning because internal components should not be handled before the system is made safe. I added a visual inspection to catch loose screws, incorrect standoffs and disconnected cables early. I also kept hardware and POST checks ahead of software checks so the technician can isolate physical assembly problems first.

ABC Solutions application: For computers that fail immediately after assembly, the early power, RAM, CPU/board compatibility and POST checks address common installation causes. For hardware-detection errors, the BIOS/UEFI verification step helps determine whether RAM or storage is being detected.

FINAL SUBMISSION CHECKLIST

- Section A: All six questions answered.
- Section B: Component table completed with actual hardware details.
- Section B: Motherboard diagram/photo labeled.
- Section B: RAM and storage removal/reinstallation documented.
- Section B: Full PC assembly steps documented with photos/diagrams.
- POST/BIOS evidence added.
- Section C: Capstone answers completed.
- Section D: Exact AI prompt included.
- AI original checklist included.
- Corrected checklist included.
- 3–4 line correction note included.
- Replace all bracketed fields before submission.

Source basis: This document follows the section structure and requirements of the uploaded CompTIA A+ AP-A1 assessment, including its requirement for practical evidence and the Section D submission items.